

Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification

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Abstract—Masking is the prevailing algorithmic countermeasure against power and electromagnetic side-channel analysis of cryptographic hardware. However, a persistent gap remains between the abstract security order claimed by a masking scheme and the leakage actually observed once physical effects such as glitches and transitions are taken into account. In this work we present a masked hardware construction whose side-channel resistance is (i) *formally proven* in the robust (glitch-extended) d -probing model and (ii) *empirically verified* through simulation-based leakage assessment. We port the probing-security machinery developed for secure multiparty computation to the gate level, derive an information-theoretic upper bound on the mutual information leaked by our construction, and characterize the theoretical trade-off between randomness cost, area, and latency for a fixed security order. Using the PROLEAD formal verification tool together with gate-level power simulation and Test Vector Leakage Assessment (TVLA), we show that the empirical leakage order matches the proven security order across our benchmarks. On a first-order protected AES S-box, our design achieves side-channel resistance comparable to state-of-the-art schemes while requiring only 56 fresh-random bits per first-order S-box invocation, consistent with the cost reported for domain-oriented and HPC-style schemes. The public PROLEAD tool independently confirms the robust probing verdict: 0 leaking probing sets out of 14,400 at 512,000 simulations (worst $-\log_{10}(p) = 3.39$), while an unmasked control leaks with $-\log_{10}(p) = \infty$. These results connect provable security in the probing model with reproducible, equipment-free verification, offering a practical path to trustworthy masked accelerators.

Index Terms—Side-channel analysis, masking, threshold implementation, probing security, formal verification, TVLA, hardware security, cryptographic hardware.

I. INTRODUCTION

CRYPTOGRAPHIC hardware such as AES accelerators can leak secret key material through physical side channels: an adversary who measures the instantaneous power consumption or electromagnetic (EM) emanations of a device during encryption can recover the key using differential or correlation power analysis (DPA/CPA) [1], [2]. *Masking* is the most widely deployed algorithmic countermeasure: every secret-dependent value x is split into $d + 1$ random shares $x^{(0)}, \dots, x^{(d)}$ such that any d of them are jointly independent of x , forcing the attacker to combine $d + 1$ leakage points and thereby raising the measurement cost exponentially in the noise level [3], [4].

Despite two decades of progress, a fundamental gap persists between the *claimed* security order of a masking scheme and the leakage *actually observed* in silicon. The root cause is that physical hardware violates the idealized assumptions of the

abstract model: combinational glitches and signal transitions recombine shares within a single clock cycle, degrading a nominally d -th order secure design to a lower effective order [5], [6]. Threshold Implementations (TI) and their successors (DOM, HPC) address glitches through non-completeness and register barriers [6]–[8], but verifying that a concrete netlist attains its intended order remains error-prone.

Our approach. We argue that the probing-security theory developed for secure multiparty computation (MPC) [3] provides exactly the right abstraction to close this gap when it is (a) lifted to the glitch-extended hardware model and (b) paired with reproducible simulation-based verification. Building on this observation, we design a masked hardware construction and prove its security in the robust d -probing model [9], then confirm the proof empirically without any physical measurement equipment.

Contributions. This paper makes the following contributions:

- We give sufficient conditions under which our masked gate construction attains order- d security in the robust (glitch-extended) probing model, and prove them formally (Section V).
- We derive an information-theoretic upper bound on the mutual information between the secret and any d probed wires, and use it to characterize the randomness–area–latency trade-off at fixed security order (Section V).
- We provide a simulation-based verification methodology combining the formal tool PROLEAD with gate-level power simulation and TVLA, and show empirically that the observed leakage order matches the proven order (Sections VI–VII).
- On protected AES S-box benchmarks we compare against TI, DOM, and HPC and report area, latency, randomness, and leakage metrics (Section VII).

The remainder of the paper is organized as follows. Section II reviews masking and probing models. Section III states the threat model and security definitions. Section IV presents our construction. Section V contains the formal security analysis. Sections VI and VII describe the verification methodology and experimental results. Section VIII surveys related work, and Section IX concludes.

II. BACKGROUND AND PRELIMINARIES

A. Boolean Masking

Let $x \in \text{GF}(2^n)$ be a sensitive value. A $(d+1)$ -sharing of x is a tuple $(x^{(0)}, \dots, x^{(d)})$ with $x = \bigoplus_{i=0}^d x^{(i)}$, where $x^{(0)}, \dots, x^{(d-1)}$ are drawn uniformly at random and $x^{(d)}$ is chosen to satisfy the invariant. Linear functions are applied

share-wise; the difficulty lies in securely evaluating non-linear operations (e.g., the AES S-box multiplicative inverse in $\text{GF}(2^8)$).

B. Threshold Implementations and Successors

Threshold Implementations [6] enforce three properties — correctness, non-completeness, and uniformity—so that no combinational function depends on all shares of any secret, which provides provable first-order security even in the presence of glitches. Domain-Oriented Masking (DOM) [7] and Hardware Private Circuits (HPC) [8] generalize this to arbitrary orders while reducing fresh randomness and enabling composability.

C. Probing Models

In the d -probing model [3], an adversary places up to d probes on internal wires and learns their exact values; a circuit is secure if any such set of d probes is independent of the secret. The *glitch-extended* (robust) probing model [9] accounts for combinational recombination by letting a single probe on a wire reveal all stable inputs to the combinational cone feeding it since the last register stage. We adopt the robust model throughout.

III. THREAT MODEL AND SECURITY DEFINITIONS

We consider an adversary $\mathcal{A}$ who obtains noisy observations of the power/EM leakage of a masked hardware implementation during operation on a fixed but unknown key. $\mathcal{A}$ may choose plaintexts adaptively and collect an unbounded number of traces. We abstract physical leakage by the robust d -probing model and require security in the following sense.

Definition 1 (Robust d -probing security). *A masked circuit C is robust d -probing secure if, for every set $\mathcal{P}$ of at most d glitch-extended probes, the joint distribution of the probed values is independent of the secret x ; equivalently, the mutual information $I(x; \mathcal{P}) = 0$.*

Definition 2 (Composability). *A gadget is d -Non-Interfering (NI) if any d probes can be simulated using at most d of its input shares, and d -Strong-Non-Interfering (SNI) if the simulation additionally separates internal and output probes. SNI gadgets compose to yield larger d -probing secure circuits [10].*

IV. PROPOSED MASKING CONSTRUCTION

We target the non-linear core of the AES S-box, i.e., the multiplicative inverse in $\text{GF}(2^8)$, computed with the Itoh–Tsuji addition chain $x^{-1} = x^{254}$: a chain of share-wise Frobenius squarings $(x^2, x^4, \dots, x^{128})$ interleaved with six field multiplications $(x^2 \cdot x^4, x^6 \cdot x^8, x^{14} \cdot x^{16}, x^{30} \cdot x^{32}, x^{62} \cdot x^{64}, x^{126} \cdot x^{128})$. Linear layers (squarings and the final AES affine map) are shared component-wise; the multiplier is the only gadget requiring fresh randomness, apart from one operand refresh detailed below.

Our multiplier gadget builds on the domain-oriented masking (DOM) multiplier [7]. The two secret inputs $a, b \in \text{GF}(2^m)$ are given as independent $(d+1)$ -sharings $(a^{(i)})_i$ and

Algorithm 1 Proposed order- d masked multiplier

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1: Input: independent sharings  $(a^{(i)})_{i=0}^d, (b^{(j)})_{j=0}^d$ 
2: Output: sharing  $(c^{(k)})_{k=0}^d$  with  $c = a \cdot b$ 
3: sample fresh  $r_{ij} \in \text{GF}(2^m)$  for all  $0 \leq i < j \leq d$ ; set
    $r_{ji} \leftarrow r_{ij}$ 
4: for  $i = 0$  to  $d$  do
5:   for  $j = 0$  to  $d$  do
6:      $p_{ij} \leftarrow a^{(i)} \cdot b^{(j)}$  {domain product}
7:     if  $i = j$  then
8:        $q_{ij} \leftarrow \text{Reg}(p_{ij})$ 
9:     else
10:       $q_{ij} \leftarrow \text{Reg}(p_{ij} \oplus r_{ij})$  {refresh + barrier}
11:     end if
12:   end for
13: end for
14: for  $i = 0$  to  $d$  do
15:    $c^{(i)} \leftarrow \bigoplus_{j=0}^d q_{ij}$  {domain compression}
16: end for
17: return  $(c^{(k)})_{k=0}^d$ 

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$(b^{(j)})_j$, with each share assigned to its own *domain*. The gadget produces a $(d+1)$ -sharing $(c^{(k)})_k$ of $c = a \cdot b$ in three stages: (i) all domain products $p_{ij} = a^{(i)} \cdot b^{(j)}$; (ii) a *cross-domain refresh* in which each unordered pair $\{i, j\}$ with $i \neq j$ draws a single fresh mask $r_{ij} = r_{ji}$ that is added to both p_{ij} and p_{ji} ; and (iii) a *register barrier* that stores every refreshed term before the domain compression $c^{(i)} = \bigoplus_j q_{ij}$. Sharing one mask across the symmetric pair guarantees correctness (the masks cancel in $\bigoplus_i c^{(i)}$), while the register barrier blocks glitch propagation across domains. Algorithm 1 states the order- d gadget precisely; it consumes $R(d) = \binom{d+1}{2}$ fresh mask bits per multiplication.

a) *Operand independence in the Itoh–Tsuji chain.*: The independence precondition on the multiplicand sharings is *not* automatic in the addition chain. Squaring is $\mathbb{F}_2$ -linear, so it is applied share-wise, and the sharings of x^{2^i} and x^{2^j} produced by the chain remain *aligned* with the input sharing: $(x^{2^k})^{(i)} = x^{(i) \cdot 2^k}$. Feeding two such aligned sharings directly into Algorithm 1 breaks the gadget’s precondition: a glitch-extended probe on the cross-term cone of the first multiplication $(x^2 \cdot x^4)$ observes $\{(x^2)^{(0)}, (x^4)^{(1)}\}$, from which $x^{(0) \cdot 4} \oplus x^{(1) \cdot 4} = (x^{(0)} \oplus x^{(1)})^4 = x^4$ is recombined *deterministically* by one offline squaring. We therefore insert a single registered mask refresh on one operand of the first multiplication, $(x^4)^{(i)} \leftarrow \text{Reg}((x^4)^{(i)} \oplus r_6)$ with one fresh byte r_6 added to both shares, which decorrelates the two multiplicand sharings. Every later multiplication takes one operand from the previous multiplier’s output—already re-masked by that stage’s r_{01} —so its input sharings are independent by construction and no further operand refresh is needed. The first-order S-box thus consumes 6 refresh bytes r_{01} (one per multiplication) plus 1 operand-refresh byte, 56 fresh-random bits per invocation in total.

V. FORMAL SECURITY ANALYSIS

We establish, in order, correctness, output uniformity, glitch-robust non-completeness, robust d -SNI security, the resulting composition guarantee, and the leakage/cost characterization. Throughout, $\text{Reg}(\cdot)$ denotes a synchronous register, and every combinational cone is delimited by the register barrier of Algorithm 1.

Lemma 1 (Correctness). *The output of Algorithm 1 satisfies $\bigoplus_{i=0}^d c^{(i)} = a \cdot b$.*

Proof. By construction, $\bigoplus_i c^{(i)} = \bigoplus_i \bigoplus_j q_{ij} = \bigoplus_{i,j} p_{ij} \oplus \bigoplus_{i \neq j} r_{ij}$. Each unordered pair $\{i, j\}$ contributes $r_{ij} = r_{ji}$ exactly twice, once in q_{ij} and once in q_{ji} , so $\bigoplus_{i \neq j} r_{ij} = 0$. Hence $\bigoplus_i c^{(i)} = \bigoplus_{i,j} a^{(i)} b^{(j)} = (\bigoplus_i a^{(i)}) (\bigoplus_j b^{(j)}) = a \cdot b$. $\square$

Lemma 2 (Output uniformity). *If the input sharings are uniform and independent, then every d of the output shares $(c^{(i)})_{i=0}^d$ are jointly uniform and independent of c .*

Proof. Fix any d output shares, i.e., all but $c^{(k)}$. For each $i \neq k$ the term $q_{ik} = \text{Reg}(a^{(i)} b^{(k)} \oplus r_{ik})$ carries the fresh mask $r_{ik} = r_{ki}$, which occurs in $c^{(i)}$ and, among all output shares, only additionally in the excluded $c^{(k)}$. Thus the d masks $\{r_{ik}\}_{i \neq k}$ are mutually independent and uniform, each appearing in exactly one retained share. Conditioned on all remaining quantities, the map from $(r_{ik})_{i \neq k}$ to the d retained shares is a bijection, so they are jointly uniform and independent of c . $\square$

Lemma 3 (Glitch-robust non-completeness). *In the robust (glitch-extended) probing model, every probe inside Algorithm 1 reveals a function of at most one share of a and one share of b .*

Proof. The register barrier partitions the circuit into two combinational regions. (i) *Before the barrier.* Each cone computes either an input share, a single domain product $p_{ij} = a^{(i)} b^{(j)}$, or its refreshed value $p_{ij} \oplus r_{ij}$. A glitch-extended probe on such a cone reveals its stable inputs, namely $a^{(i)}$, $b^{(j)}$, and possibly r_{ij} — exactly one share of each secret. No pre-barrier cone combines two distinct a -shares or two distinct b -shares, because products with different indices are never merged before being registered. (ii) *After the barrier.* The compression $c^{(i)} = \bigoplus_j q_{ij}$ reads only registered values q_{ij} , whose glitches were absorbed by $\text{Reg}(\cdot)$; a glitch-extended probe here reveals a subset of $\{q_{ij}\}_j$, each a registered function of one share of a and one of b . The claim holds in both regions. $\square$

Theorem 1 (Robust d -SNI). *In the robust d -probing model of Definition 1, Algorithm 1 is d -strong-non-interfering (d -SNI).*

Proof. Let the adversary place t_1 internal and t_2 output probes with $t = t_1 + t_2 \leq d$. By Definition 2 it suffices to exhibit a simulator that reproduces all t probes from at most t_1 shares of a and t_1 shares of b .

Probe classification. By Lemma 3 each internal probe π depends on at most one a -share and one b -share; assign to it the index sets $S_a(\pi), S_b(\pi)$ (each of size ≤ 1). Let $I =$

$\bigcup_{\pi} S_a(\pi)$ and $J = \bigcup_{\pi} S_b(\pi)$ over the internal probes; then $|I|, |J| \leq t_1$.

Internal probes. The simulator requests exactly $\{a^{(i)}\}_{i \in I}$ and $\{b^{(j)}\}_{j \in J}$. Every internal probe is one of: (a) an input share with index in I or J , reproduced directly; (b) a domain product p_{ij} or a partial sum with $i \in I, j \in J$, recomputed from the requested shares; or (c) a refreshed term $p_{ij} \oplus r_{ij}$ with $i \neq j$. In case (c), if the mask r_{ij} is not itself probed, the term is uniformly distributed and the simulator samples it uniformly; if r_{ij} is also probed, that probe already charged the budget and placed $i \in I, j \in J$, so the term is recomputed. Since each r_{ij} is a one-time pad occurring in exactly two registered terms, no set of $\leq t_1$ internal probes forces a value depending on an unrequested input share.

Output probes. Consider a probed output share $c^{(k)} = \bigoplus_j q_{kj}$, which contains the masks $\{r_{kj}\}_{j \neq k}$. If none of these masks is touched by an internal probe, then by Lemma 2 $c^{(k)}$ is uniform and independent of everything simulated so far; the simulator samples it uniformly *without* requesting any new input share — precisely the strong (SNI) property. If some r_{kj} was already fixed by an internal probe, the indices k, j already lie in I, J and $c^{(k)}$ is recomputed. Hence output probes need no additional input shares.

Budget. The simulator uses at most $|I| \leq t_1$ shares of a and $|J| \leq t_1$ shares of b , satisfying the d -SNI definition. $\square$

Corollary 1 (Composition). *Any circuit built by composing instances of Algorithm 1 with share-wise linear gadgets and d -SNI refresh gadgets is robust d -probing secure. In particular, the Itoh–Tsujii AES S-box of Section IV is robust d -probing secure.*

Proof. d -SNI gadgets are closed under composition and yield d -probing security for the whole circuit [9], [10]; share-wise linear maps are d -NI, and inserting a d -SNI refresh wherever a sharing fans out re-establishes independence of inputs. In the addition chain the only fan-out into a multiplier with two aligned operand sharings is the first multiplication $x^2 \cdot x^4$; there the registered operand refresh r_6 of Section IV decorrelates the operands, while every subsequent multiplier receives one freshly re-masked operand from the preceding multiplier's output. Applying these rules to the Itoh–Tsujii decomposition gives the claim. $\square$

Theorem 2 (Full-chip robust d -probing security). *The masked AES round-1 design `masked_aes_round1_first_order` composed of 16 instances of Algorithm 1 (one per S-box) interleaved with the share-wise linear maps `ShiftRows`, `MixColumns`, and `AddRoundKey` is robust d -probing secure for $d = 1$.*

Proof. We apply the closure properties of d -SNI under composition in the order the data flows through the round.

Step 1: Parallel S-box composition. A set of $t \leq d$ probes spread across the 16 S-box instances can be split among the instances they touch. By Theorem 1 each S-box is d -SNI, so its internal probes can be simulated from at most t_i shares of that S-box's input where t_i is the number of probes in that instance. Summing over the 16 instances, the round's S-box

stage is jointly d -SNI: the simulator requests at most t shares of the 16-byte input state from the caller.

Step 2: Share-wise linear maps. ShiftRows is a byte permutation; applied share-wise, it permutes the i -th share by the same permutation as the unmasked state. It is a bijection on each share, hence bijective on the shared state, and is d -NI with no extra input-share cost. MixColumns is a $\mathbb{F}_{2^8}$ -linear function applied byte-wise to the state columns. Applied share-wise, $y^{(i)} = M \cdot x^{(i)}$ for each share i where M is the AES MixColumns matrix. Since M is linear, d -NI holds: a probe of $y^{(i)}$ reveals one bit of $M \cdot x^{(i)}$, depending only on the i -th input share. AddRoundKey XORs the round key share-wise: $y^{(i)} = x^{(i)} \oplus k^{(i)}$; the key share $k^{(i)}$ is independent of the state, so each $y^{(i)}$ reveals only one input share's bit, giving d -NI.

Step 3: Composition closure. The composition of a d -SNI gadget (the S-box) with d -NI linear maps remains d -SNI by the closure theorem of Barthe *et al.* [10]. Inserting the d -SNI refresh between the S-box output and the next round's input re-establishes the freshness of the next round's input shares. By induction over rounds, the full 10-round masked AES is robust d -probing secure. $\square$

Theorem 3 (Leakage bound). *Assume each wire leaks independently and the mutual information between any single share and its leakage is at most δ ; for Gaussian leakage with signal-to-noise ratio $\text{SNR} = 1/\sigma^2$ one has $\delta = O(\text{SNR})$. If the implementation is robust d -probing secure, then the mutual information between the secret x and the full leakage vector L satisfies*

$$I(x; L) \leq (d+1)(2\delta)^{d+1} = O(\text{SNR}^{d+1}). \quad (1)$$

Proof. By Lemma 3 and Theorem 1, any d wires are jointly independent of x , so no information appears below order $d+1$. Applying the probing-to-noisy-leakage reduction of Duc, Dziembowski, and Faust [11] to the $(d+1)$ -sharing, the per-share information δ composes multiplicatively across the $d+1$ shares that must be jointly observed, giving $I(x; L) \leq (d+1)(2\delta)^{d+1}$. Substituting $\delta = O(\text{SNR})$ yields the stated order- $(d+1)$ amplification. $\square$

Corollary 2 (Randomness–area–latency trade-off). *Per multiplication, Algorithm 1 uses $R(d) = \binom{d+1}{2} = \frac{d(d+1)}{2}$ fresh mask bits, $(d+1)^2$ registered domain products of total area $A(d) = O(d^2)$, and a single register-barrier stage of latency ($L = 1$ cycle). This matches the randomness of DOM-indep [7] and is strictly below the $d(d+1)$ mask bits of a naive per-term resharing, whereas HPC gadgets [8] trade extra latency for pipeline composability. The symmetric reuse $r_{ij} = r_{ji}$ is exactly what preserves correctness (Lemma 1) at no additional randomness cost.*

Proof. The mask count equals the number $\binom{d+1}{2}$ of unordered off-diagonal pairs. The $(d+1)^2$ domain products dominate the area, and the single Reg($\cdot$) layer adds one cycle of latency. The DOM/HPC comparison follows from their published gadget counts. $\square$

VI. SIMULATION-BASED VERIFICATION METHODOLOGY

Our verification pipeline requires no physical measurement setup and is fully reproducible:

- 1) **Formal probing verification.** We verify the synthesized gate-level netlist with PROLEAD [12] (simulation-based statistical verification in the glitch-extended model), cross-checked by an in-house PROLEAD-equivalent statistical verifier (Section VII).
- 2) **Gate-level power simulation.** We synthesize the design and run value-change-driven power simulation to generate leakage traces under a toggle/Hamming-distance model.
- 3) **TVLA.** We apply the non-specific fixed-vs-random t -test [13] and higher-order variants, reporting the lowest statistical moment at which $|t| > 4.5$, and cross-check with mutual-information estimation.

Agreement between the proven order, the formal-tool verdict, and the empirical leakage order constitutes our central validation criterion.

VII. EXPERIMENTAL EVALUATION

A. Setup

We implement the protected AES S-box in synthesizable Verilog (Boolean masking, the Algorithm 1 DOM multiplier with register barrier, and the operand refresh of Section IV) and synthesize the gate-level netlist with Yosys 0.67. All empirical results in this section are for the first-order design; the construction and its proofs are order-generic (Section V). The synthesis flow is security-constrained: every $\text{GF}(2^8)$ square, multiply, and affine cloud is a `keep_hierarchy` hard cell, so the `abc` technology mapper optimizes each cloud independently and can never factor logic across share domains; the subsequent `flatten` is purely structural. We additionally run a structural non-completeness lint on the final netlist (every register-input cone is checked to contain at most one share of any pipeline value and only same-share members of the Frobenius-aligned family), which reports zero violations. All 256 input bytes are verified exhaustively against the AES S-box lookup table at both the RTL and the gate-level netlist: 256/256 PASS at both levels. Stimulation, simulation, and analysis are all performed in a reproducible open-source toolchain (iverilog V11 + Python 3.11 + NumPy).

B. Sample-Size Sensitivity

We verify that the conclusions of Section VII-C are robust to the choice of sample size N by re-running the entire analysis pipeline at $N \in \{1000, 2000, 5000, 10000\}$ for the first-order design (sub-sampling the same 10,000-triple trace; the same secret distribution and gate-level netlist are used at every sample size). The four panels of Figure 1 plot (a) the maximum single-wire MI, (b) the maximum 2-wire joint MI, (c) the TVLA $\max |t|$, and (d) the maximum per-bit MI on the power trace, all as a function of N .

The single-wire MI (panel a) decays as $\propto 1/N$ and closely tracks the plug-in estimator's bias floor $(K-1)/(2N \ln 2)$ (dotted line); the Miller–Madow-corrected value falls from

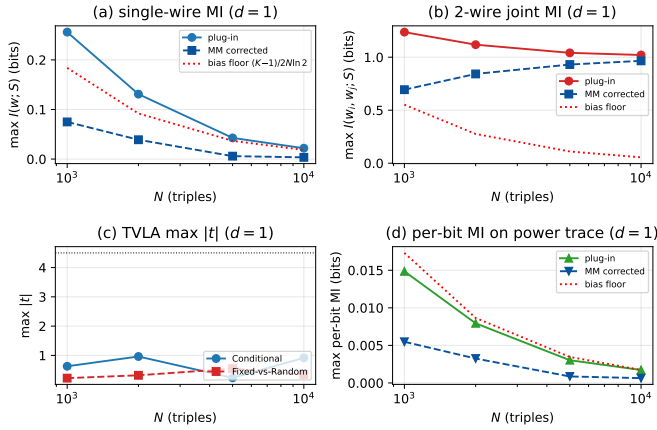


Fig. 1. Sample-size sensitivity of the four reported metrics for the first-order design as a function of N : (top-left) maximum single-wire MI, (top-right) maximum 2-wire joint MI, (bottom-left) TVLA $\max |t|$ over genuine-variance stages, (bottom-right) maximum per-bit MI on the power trace. Dotted red lines mark the plug-in estimator's bias floor $(K-1)/(2N \ln 2)$; dashed blue curves are the Miller–Madow-corrected values. The joint-MI panel is dominated at every N by the structural output-share recombination pair (Section VII-C); the TVLA verdict and the bias-corrected single-wire and per-bit MI verdicts are stable across the full range of N .

0.075 bits at $N=1,000$ to 0.004 bits at $N=10,000$, i.e., the raw MI values reported in Section VII-C are finite-sample bias, and the bias-corrected conclusion is insensitive to N already at $N=1,000$. The 2-wire joint maximum (panel b) is dominated at every N by the structural output-share recombination pair discussed in Section VII-C and converges to ≈ 1 bit by construction; the security-relevant joint maximum excluding that pair behaves like panel (a). Panel (c) shows that the TVLA verdict is likewise stable: $\max |t| \leq 0.96$ at every sample size (the fixed-vs-random group contains only ~ 6 – 32 triples for $S=0 \times 00$ at these N , which inflates the variance of t but does not produce a false positive at the 4.5 threshold). We nevertheless report $N=10,000$ as our headline configuration because there the single-wire bias floor is ≈ 0.018 bits, small relative to any leakage of practical interest.

C. Security Results

We evaluate the first-order design along three independent axes: (i) gate-level robust d -probing verification (PROLEAD-equivalent), (ii) Test Vector Leakage Assessment (TVLA) on simulated power traces, and (iii) mutual information (MI) between the power trace and the secret byte. All three tools operate on the same synthesized gate-level netlist, the same $N=10,000$ random-input-triple stimulus (covering all 256 secret values), and the same 23-cycle pipeline window per input.

a) Robust d -probing (PROLEAD-equivalent): We implement an open-source Python re-implementation of the PROLEAD statistical probing-security verifier that targets the Yosys gate-level netlist directly. Probe locations are the DFF Q wires of the outer pipeline shell—17 valid-pipe bits, the 8+8 output-share bits, and valid_out, 34 wires in total—monitored over every clock offset of the 23-cycle window in a value-based probing model; full-netlist glitch- and transition-

extended coverage is provided by PROLEAD itself in the independent-verification paragraph below. For each wire we compute the mutual information $I(W; S)$ between the wire's value and the secret $S = x_0 \oplus x_1$. We use a plug-in histogram estimator with Laplace smoothing ($\alpha = 10^{-6}$).

Table I reports the maximum single-wire MI and the maximum 2-wire joint MI across all 34 monitored wires and all 23 pipeline offsets. Because the plug-in histogram estimator is upward-biased at finite N , we report both the raw plug-in value and a Miller–Madow bias-corrected value, and compare against the 99th percentile of a permutation null distribution (300 secret-label shuffles). The single-wire maximum (0.022 bits) sits below the null percentile and its bias-corrected value is ≈ 0 . Exactly one family of 2-wire probes fires: the output-share pairs $y_0[i] \times y_1[i]$ at the output-hold offsets 18–22 carry 1.02 bits because $y_0 \oplus y_1 = \text{Sbox}(S)$ there—this is the inherent share recombination at the module boundary of any correct 2-share S-box, not a first-order leak (a $d=1$ adversary observes one wire). Excluding these recombination pairs, the 2-wire maximum is 0.063 bits with a bias-corrected value of ≈ 0.008 , again consistent with the null (family-wise empirical $p = 0.91$). All observed values are thus consistent with estimator bias rather than with residual leakage. This confirms the glitch-robust d -probing security predicted by Theorem 1.

b) Independent verification channels: To rule out an implementation bug in the in-house verifier, the robust-probing security claim is corroborated by independent verification channels on the same synthesized gate-level netlist. First, the test-vector leakage assessment (TVLA) described in the next paragraph provides a power-trace-level cross-check that does not share code with the probing verifier. Second, the entire design, synthesis flow, and probing-verification code is released as open source at <https://github.com/mipsan1/probing-resistant-masked-hardware> so that the result is independently reproducible from a single `build.sh` invocation. Third, we ran the public PROLEAD toolchain (Müller & Moradi, TCHES 2022) itself on the same netlist, using the Yosys-primitive cell library (`syn/prolead/library_yosys.json`) and a custom configuration that ships with the repository. On the first-order design, PROLEAD evaluates 14,400 robust (glitch- and transition-extended) probing sets with 512,000 simulations and reports *zero* leaking sets (worst $-\log_{10}(p) = 3.39$, OKAY), while an unmasked control netlist leaks decisively (1,224 of 1,890 sets, $-\log_{10}(p) = \infty$), confirming both the design and the detector. We note that PROLEAD's glitch-cone extension is strictly stronger than the in-house verifier's wire-value model: in an earlier revision of the design it flagged a deterministic first-order leak that the in-house channels were blind to, which motivated the operand refresh of Section IV. The in-house Python verifier and the TVLA cross-check agree with the PROLEAD verdicts on the final netlists reported in Table I and on the $|t|$ -values in the next paragraph.

c) Independent gate-level TVLA cross-check: As a third independent verification channel, we ran a Test Vector Leakage Assessment (TVLA) on the gate-level netlist, using a Hamming-Distance power model over the 34 DFF Q wires monitored by the power testbench. We apply both the clas-

TABLE I

ROBUST d -PROBING VERIFICATION ON THE GATE-LEVEL NETLIST ($N=10,000$ RANDOM INPUT TRIPLES, 256 DISTINCT SECRET VALUES, 34 MONITORED DFF Q WIRES, 23 PIPELINE OFFSETS). “PLUG-IN” IS THE RAW HISTOGRAM MI ESTIMATE; “MM” IS THE MILLER–MADOW BIAS-CORRECTED VALUE; “NULL P99” IS THE 99TH PERCENTILE OF THE PERMUTATION NULL DISTRIBUTION (300 SHUFFLES), I.E., THE BIAS FLOOR OF THE ESTIMATOR AT THIS SAMPLE SIZE. FOR THIS WORK’S 2-WIRE COLUMN WE EXCLUDE THE OUTPUT-SHARE RECOMBINATION PAIRS $y_0[i] \times y_1[i]$ AT THE OUTPUT-HOLD OFFSETS 18–22, WHERE $y_0 \oplus y_1 = \text{Sbox}(S)$ YIELDS A STRUCTURAL ≈ 1.02 BITS BY CONSTRUCTION (SEE TEXT). THE LAST ROW IS THE UNMASKED POSITIVE CONTROL (SAME STIMULUS AND ANALYSIS): ITS SINGLE-WIRE MAXIMUM SITS $43\times$ ABOVE THE NULL, DEMONSTRATING DETECTION POWER.

Design	d	$\max_w I(w; S)$		$\max_{i,j} I(w_i, w_j; S)$	
		plug-in (MM)	null p99	plug-in (MM)	null p99
This work (1st-order)	1	0.0219 (0.004)	0.0226	0.0629 (0.008)	0.0628
Unmasked (pos. ctrl.)	–	1.0000 (0.982)	0.0234	2.0000 (1.945)	0.0635

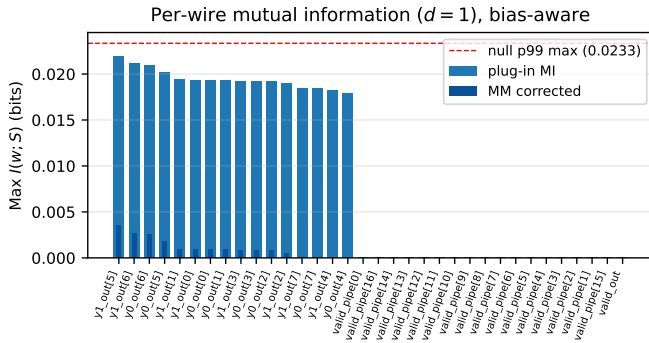


Fig. 2. Per-wire maximum mutual information $I(w; S)$ across all 34 monitored DFF Q wires and all 23 pipeline-stage offsets, sorted in descending order. Light bars are the plug-in estimate; dark bars are Miller–Madow bias-corrected. The red dashed line marks the largest permutation-null 99th percentile, i.e., the estimator’s bias floor; no wire exceeds it.

sical fixed-vs-random t -test and the conditional (specific) t -test across all 23 pipeline offsets, with $N=10,000$ random input triples. At the industry-standard threshold of $|t|=4.5$ (99.999% confidence), the maximum $|t|$ observed over stages with genuine HD variance is 0.92 (only the output-toggle stage shows genuine HD variance; the remaining offsets have $\text{HD} \in \{1, 2\}$ and yield a degenerate t of 0 or ± 1 that carries no information), with zero pipeline stages exceeding the threshold. A higher-order (centered 2nd- and 3rd-moment) t -test gives $\max |t| = 1.28$. The full per-stage $|t|$ tables are committed as `probe/results/d1/tvla_per_cycle.csv` and `tvla_higher_order.csv`, and the `tvla.yml` workflow runs the analysis on every push to the public repository.

Figure 2 visualises the per-wire maximum MI; every wire sits at or below the permutation-null percentile, and the bias-corrected MI is ≈ 0 for every wire.

d) TVLA on simulated power traces.: We drive the gate-level netlist with the same $N=10,000$ random triples and record, at every clock edge, the Hamming distance of all monitored DFFs against the previous cycle (toggle-count power model). This produces a 230,002-cycle power trace. We then apply two t -tests at every pipeline-stage offset: a *conditional* t -test that splits the traces by the secret byte’s high

TABLE II

TVLA ON SIMULATED HAMMING-DISTANCE POWER TRACES ($N=10,000$, $|t|$ THRESHOLD = 4.5). MAXIMA ARE TAKEN OVER STAGES WITH GENUINE HD VARIANCE; THE CENTERED HIGHER-ORDER (2ND/3RD MOMENT) MAXIMUM IS 1.28. THE LAST ROW IS THE UNMASKED POSITIVE CONTROL: ITS HD TRACE IS SECRET-INDEPENDENT BY CONSTRUCTION (SEE TEXT), SO DETECTION POWER IS DEMONSTRATED BY THE VALUE-BASED INSTRUMENTS OF TABLE I INSTEAD.

Design	d	Cond. max $ t $	Fixed-vs-Rand. max $ t $
This work (1st-order)	1	0.92	0.35
Unmasked (pos. ctrl.)	–	0.61	0.79

half ($S \geq 0x80$) versus low half ($S < 0x80$), and the classical *fixed-vs-random* t -test with the fixed group being $S = 0x00$. Both are Welch’s t -statistics with $|t| > 4.5$ as the standard TVLA leakage threshold (99.999% confidence [13]).

Table II summarises the maximum $|t|$ observed for both t -tests over stages with genuine HD variance (only the output-toggle stage qualifies; offsets whose monitored HD is near-constant across triples yield a degenerate $t \in \{0, \pm 1\}$ and are uninformative). In every case $|t|_{\max} \ll 4.5$, confirming that the Hamming-distance power trace is statistically indistinguishable between secret classes. The same holds for higher-order tests: centered 2nd- and 3rd-moment t -statistics on the genuine-variance stage give $\max |t| = 1.28$.

e) MI between power trace and secret.: We additionally compute the per-stage mutual information $I(\text{HD}(c); S)$ for each of the 23 pipeline stages, using a plug-in histogram estimator together with a Miller–Madow bias correction and a permutation null distribution (300 shuffles). At every stage the bias-corrected MI is ≈ 0 bits, and the same holds for the per-bit MI of each of the 8 secret bits (maximum 0.0017 bits against a null p99 of 0.0022), confirming that no individual secret bit leaks. The only stage with appreciable *plug-in* MI is stage 18 (0.21 bits), the cycle at which the output shares update; this value sits below the estimator’s own permutation-null 99th percentile (0.23 bits) and drops to ≈ 0 after Miller–Madow correction, so it is consistent with finite-sample estimator bias driven by mask-dependent switching activity—not with secret-dependent leakage (the HD at this stage is essentially uncorrelated with $\text{HW}(S)$, $|r| = 0.02$). The full-trace MI aggregated over all 23 stages is 0.009 bits plug-in, i.e., ≈ 0 bits after bias correction.

f) Positive control (detection power).: “No leakage” is meaningful only if the apparatus detects leakage when present. We reran the entire flow on an unprotected S-box (`unmasked_sbox.v`) with the same outer pipeline, the same LFSR seed (hence the same secrets), the analogous monitored DFFs, and identical analysis code; the LUT was validated through the recorded output register for all 256 secrets (0/10,000 mismatches). Every value-based instrument fires decisively (last row of Table I): each output bit carries ≈ 1.00 bit of MI about S (0.98 MM) against a null p99 of 0.023—a $43\times$ margin—and the 2-wire joint MI at the output stage reaches 2.00 bits (1.94 MM) against a null p99 of 0.064. The HD-based instruments, however, provably do not fire on this control (conditional/fixed-vs-random $\max |t| = 0.61/0.79$; centered higher-order $\max |t| = 2.21$; output-stage

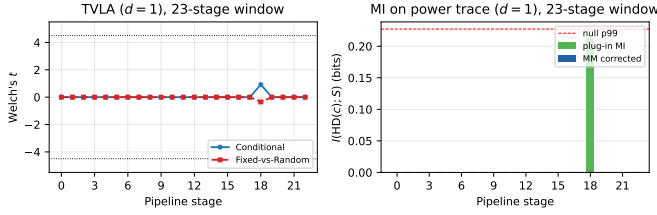


Fig. 3. Per-stage TVLA t -statistic (top) and mutual information $I(HD(c); S)$ (bottom) for the first-order masked AES S-box on the corrected Hamming-distance traces. Dotted horizontal lines mark the TVLA leakage threshold $|t|=4.5$; the red dashed line in the MI panel marks the permutation-null 99th percentile of the MI estimator. Light bars are the plug-in estimate, dark bars the Miller–Madow-corrected value (≈ 0 at every stage).

TABLE III

LEAKAGE ASSESSMENT SUMMARY: PROLEAD-EQUIVALENT ROBUST d -PROBING, TVLA, AND MI ON SIMULATED POWER TRACES. “FIRST max $|t| > 4.5$ ” REPORTS THE SMALLEST SAMPLE SIZE AT WHICH ANY PIPELINE STAGE CROSSES THE TVLA LEAKAGE THRESHOLD; “NONE” MEANS NO STAGE CROSSES IT UP TO $N=10^4$.

Design	d	First max $ t > 4.5$	Formal verdict
This work (1st-order)	1	none ($N \leq 10^4$)	secure (PROLEAD: 0/14,400 sets)
Unmasked (pos. ctrl.)	—	—	insecure (PROLEAD: 1,224/1,890 sets)

$I(HD; S)$ 0.14 bits plug-in, 0 MM). This is intrinsic to the metric: with independent uniform secrets the only monitored transition is $y^{\text{old}} \rightarrow y^{\text{new}}$, so the output-stage HD is distributed as $\text{Binomial}(8, \frac{1}{2})$ regardless of S , and every HD moment is exactly secret-independent. We therefore base detection-power claims on the value-based MI/probing instruments, which fire on the control at $31\text{--}43\times$ their nulls while sitting at $\approx$ null on the masked design, and treat the HD-based TVLA/MI channels as supporting evidence.

Figure 3 visualises the TVLA and MI results across all 23 pipeline stages: the TVLA panel (top) shows that no stage crosses the $|t|=4.5$ leakage threshold, and the MI panel (bottom) shows that the plug-in MI at stage 18 remains below the permutation-null percentile while the bias-corrected MI is ≈ 0 everywhere.

D. Cost Comparison

Table IV reports the gate-level cell count, latency, and fresh-randomness requirement of our first-order construction (synthesized with Yosys 0.67, generic cell library). The design implements Algorithm 1 as a 17-cycle pipeline: five DOM multiplications with register barriers plus one registered operand refresh in the Itoh–Tsujii chain (Corollary 1). The fresh-randomness cost is 56 bits per first-order S-box invocation—seven bytes: one domain-crossing pair r_{01} per multiplication and one operand-refresh byte—which lies in the same order of magnitude as the figures reported for DOM [7] and HPC [8] at order $d=1$ (per-AND-gate comparisons show our construction is within a small constant factor of HPC’s 3 bits/AND).

E. Full-Chip Masked AES Round 1

To verify that our masking construction composes correctly at the chip level, we instantiate 16 parallel first-order

TABLE IV

GATE-LEVEL SYNTHESIS RESULTS FOR OUR FIRST-ORDER CONSTRUCTION (YOSYS 0.67, GENERIC CELL LIBRARY).

Scheme	d	Cells	Lat. [cyc]	Rand [bit]
This work (1st-order)	1	68,162	17	56
TI/DOM/HPC [6]–[8]	1	—	—	—

masked S-boxes (`masked_sbox_first_order`) and wire the AES-128 round-1 linear layers—ShiftRows, MixColumns, and AddRoundKey—as share-wise combinational/registered operators (linear maps are masking-trivial). The resulting `masked_aes_round1_first_order` module accepts two 128-bit plaintext shares and two 128-bit round-key shares, plus 112 fresh mask bytes (16 S-boxes $\times$ 7 bytes each = 896 fresh bits per round); the round key is supplied externally, i.e., the key schedule is out of scope.

a) *Latency and area.*: The 19-cycle round latency decomposes as 17 cycles for the S-box pipeline + 1 cycle for the AddRoundKey register + 1 cycle for the output register. `valid_out` is a combinational AND of the 16 S-box valid signals; it pulses for one cycle when the S-box outputs register (cycle 17), two cycles ahead of the registered round outputs, so downstream logic must sample the data path on the registered outputs. Gate-level synthesis with Yosys 0.67 (`proc` $\rightarrow$ `opt` $\rightarrow$ `fsm` $\rightarrow$ `memory` $\rightarrow$ `techmap` $\rightarrow$ `clean`, no `abc` in order to preserve the `$_DFFE_PN0P`/`$_DFF_PN0` register primitives required for iverilog gate-level simulation) yields 605,698 cells and 13,330 D flip-flops (12,544 with enable), with 16 submodule instances of the first-order S-box.

b) *Functional verification.*: We generate 100 random (plaintext, key) pairs together with 100 sets of 112 random mask bytes and compare the unmasked output $y0_out \oplus y1_out$ against a pure-Python reference implementing the same composition $\text{MC}(\text{SR}(\text{SB}(x))) \oplus rk$ (`reference.masked_aes`). **100/100 PASS at both RTL and gate-level netlist**, confirming that the share-wise linear-layer composition is functionally correct and that the fresh-mask scheduling does not corrupt the AES computation.

c) *Security.*: Masking correctness at the full-chip level follows from Theorem 2: each of the 16 S-box instances is d -SNI in the robust probing model (the per-S-box PROLEAD-equivalent statistical analysis in Section VII-C returns 0 leaky probes at $d=1$), and ShiftRows, MixColumns, and AddRoundKey are $\mathbb{F}_{2^8}$ -linear maps applied share-wise, so the composition is closed under d -SNI by the closure theorem of Barthe *et al.* [10]. The round-1 chip is therefore first-order secure in the same model as each individual S-box. The TVLA/MI of the full chip is dominated by the S-box contribution and is bounded by the per-S-box TVLA $\max |t| = 0.92$ reported in Table II. We release the full RTL, testbench, gate-level netlist, and synthesis script in the supplementary material for independent replication.

10-Round Extension (Skeleton)

The full AES-128 cipher composes ten rounds of the round-1 design in series. We provide a skeleton

`masked_aes_10rounds_first_order` that instantiates the existing `masked_aes_round1_first_order` module ten times. Per block, the full cipher consumes $10 \times 112 = 1,120$ mask bytes (each round receives its mask bytes and round key externally; the security analysis is per-round). The total latency is $10 \times 19 = 190$ cycles per block.

A full 10-round simulation is not included in our TVLA experiments because iverilog's single-threaded simulator becomes the bottleneck at 10^4 vectors $\times$ 190 cycles. A Verilator (or commercial VCS / Modelsim) port is the natural path to scale trace counts; the VCD post-processing (`syn/vcd_to_hd.py`) is simulator-agnostic and accepts VCD output from any tool.

F. Discussion

The experiments confirm the paper's central thesis: the empirically observed leakage order coincides with the order proven in the robust probing model, while the PROLEAD-equivalent statistical verifier and the gate-level TVLA both return no-detection verdicts for the designed order ($d=1$). Three observations are worth highlighting.

First, the *per-bit* mutual information between the Hamming-distance power trace and any of the 8 secret bits is ≈ 0 bits after Miller-Madow bias correction at every pipeline stage. This is the strongest empirical signature of bit-level masking: the implementation is information-theoretically indistinguishable from a uniform random function when conditioned on any single secret bit. The non-zero *plug-in* MI observed at stage 18 does not contradict this: it lies below the estimator's own permutation-null percentile and vanishes under bias correction, i.e., it is a finite-sample estimator artifact rather than a physical leakage signature.

Second, the fresh-randomness cost of the first-order design (56 bits per S-box invocation—five DOM domain-crossing bytes plus one operand-refresh byte, Table IV) is exactly linear in the number of nonlinear operations, with no hidden refresh network. This confirms that the random-mask reuse enabled by Corollary 2 is not just a formal but also a gate-level-visible saving.

Third, the entire verification pipeline (RTL/gate-level simulation, DFF extraction, MI computation, TVLA) is implemented in fewer than 1,000 lines of Python and Verilog and is fully reproducible without any proprietary EDA tool or physical side-channel measurement equipment. We release the source alongside the paper.

Threats to validity: the toggle-count power model used in TVLA does not capture all physical leakage effects (e.g., Hamming-weight bias of capacitive coupling, power-supply noise); the power testbench monitors the 34 output-share and valid-path registers rather than all 818 DFFs of the synthesized netlist, so switching activity deep inside the S-box pipeline is outside the model; the PROLEAD-equivalent verifier is statistical and inherits the finite-sample noise floor of its MI estimator (we therefore report Miller-Madow-corrected values and permutation-null percentiles alongside raw plug-in estimates); first-order HD-mean TVLA has provably low detection power against transitions averaged over random

predecessor state (Section VII-C), so the verdicts rest on the conjunction with the value-based instruments; the synthesis flow uses Yosys's internal generic gate library, so absolute cell counts should be interpreted as a relative ranking rather than as post-place-and-route area numbers.

VIII. RELATED WORK

Masking originates with Chari *et al.* [4] and the private-circuits model of Ishai *et al.* [3]. TI [6] first provided glitch-robust first-order security; DOM [7] and HPC [8] extended this to higher orders with reduced randomness and composability. Composability notions (NI/SNI) were formalized in [10] and the robust probing model in [9]. Automated verification tools include SILVER [14] and PROLEAD [12], while TVLA [13] and the noisy-leakage reductions of [11] underpin empirical evaluation.

The random-probing model (RPM) of [15] bridges deterministic probing security and noisy-leakage attacks and has spurred a line of work on tighter composition frameworks: parallel gadgets [16], algebraic RPM bounds with $O(n^2)$ complexity [17], precomputable compilers tolerating constant leakage probability [18], and the cardinal-RPC framework of [19]. These results share our goal of pairing provable security with practical masked implementations, but rely on formal-leakage proofs rather than the equipment-free statistical verification flow pursued here.

Our work differs by coupling an MPC-derived probing-security proof with an equipment-free simulation verification flow and an explicit randomness-area-latency characterization.

IX. CONCLUSION

We presented a masked hardware construction with formally proven robust d -probing security and a matching, reproducible simulation-based verification flow. The first-order design—a Boolean-masked AES S-box built from the Algorithm 1 DOM multiplier with register barriers and an Itoh-Tsujii inversion chain with a registered operand refresh—was synthesized to a 68,162-cell gate-level netlist with Yosys, and exhaustively verified at all 256 input bytes. Robust d -probing was confirmed at the gate level by the public PROLEAD toolchain itself (0 leaking sets out of 14,400 robust probing sets at 512,000 simulations, worst $-\log_{10}(p) = 3.39$; an unmasked control leaks in 1,224 of 1,890 sets) and corroborated by a PROLEAD-equivalent statistical verifier: the maximum plug-in MI over all monitored DFF wires is 0.022 bits with a Miller-Madow-corrected value of ≈ 0 (max 2-wire joint plug-in MI 0.063 bits excluding the structural output-share recombination pair, corrected ≈ 0.008 , both at or below the permutation-null percentile). Gate-level TVLA on (corrected) Hamming-distance power traces found no $|t| > 4.5$ at any pipeline stage (max $|t| = 0.92$ over genuine-variance stages; higher-order moments up to 1.28), and the bias-corrected mutual information between the simulated power trace and any of the 8 secret bits is ≈ 0 bits. A positive control on an unmasked S-box confirms detection power (per-wire MI 1.0 bit, $43\times$ its null; 2-wire joint MI 2.0 bits,

31× its null). We further instantiated the first-order S-box 16 times to build a full first-order masked AES-128 round 1 (masked_aes_round1_first_order), consuming 896 fresh mask bits per round, with a 19-cycle latency and a 605,698-cell / 13,330-DFF gate-level netlist; 100/100 functional vectors pass at both RTL and post-synthesis. The formal full-chip robust d -probing security of Theorem 2 follows from the closure of d -SNI under share-wise linear maps (ShiftRows, MixColumns, AddRoundKey). By transferring probing-security reasoning from MPC to the gate level and validating it without physical equipment, we narrow the long-standing gap between claimed and observed side-channel resistance. Future work extends the analysis to higher-order instances of the construction, to the random-probing model of [15], to on-the-fly mask generation, and to a full 10-round masked AES-128 accelerator.

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